VPL: A Novel Cost Model for Earth and Mars-Orbit Navigation and Communication Constellations

M. Cholevas (TU München) • G. Vardanikas (TU Delft)

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The problem

Limited public unit-cost data for navigation and communication constellations. Legacy models (SMAD USCM8, QuickCost) rest on data up to the early 2000s. We build a cost model on **25 systems** (1997–2025), that also covers New-Space systems.

The resolved CER model

$$COST(M,N) = F \cdot g \cdot [\underbrace{DEV(M)}_{5.99 \cdot M^{0.613}} + \underbrace{PFM(M)}_{0.219 \cdot M^{0.828}} \cdot N^{LRE}]$$

- M = platform mass [kg],
- N = units, cost in FY2025 €M.
- $LRE = 1 + \ln(0.9)/\ln(2) = 0.848$
- g = 1 traditional, 0.118 New-Space.
- F = 1 Earth, 1.07 Mars.

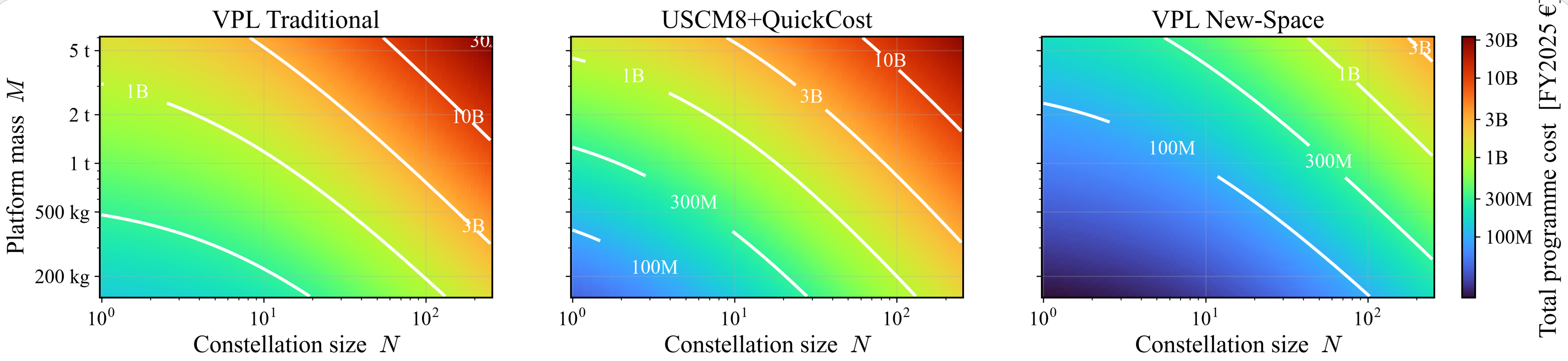
Main metrics

0.94 cross-validated R^2

5–21× New-Space cheaper

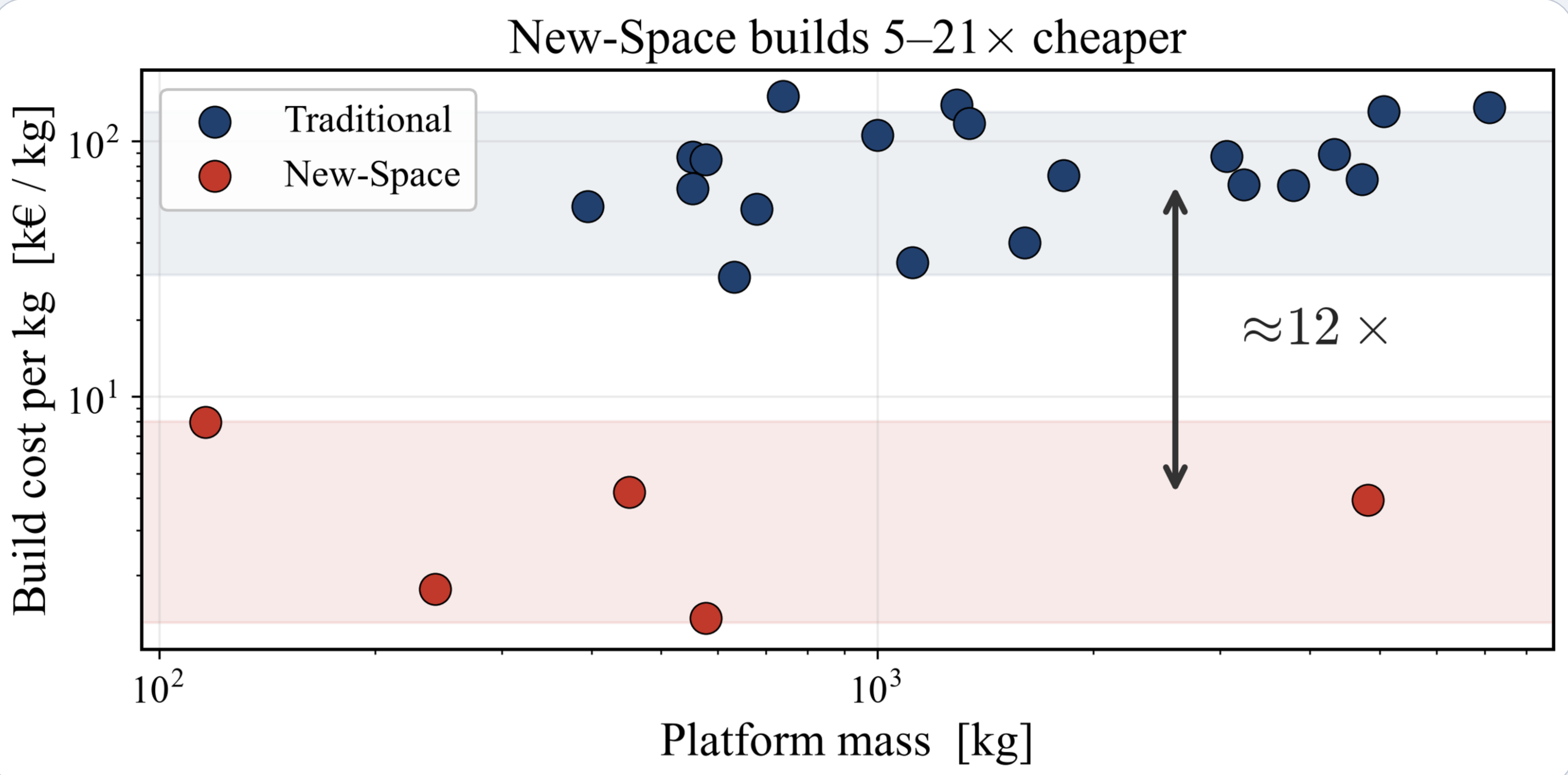
≈ USCM8+QuickCost on traditional; but SMAD over-predicts New-Space ~7×

Cost vs mass and constellation size



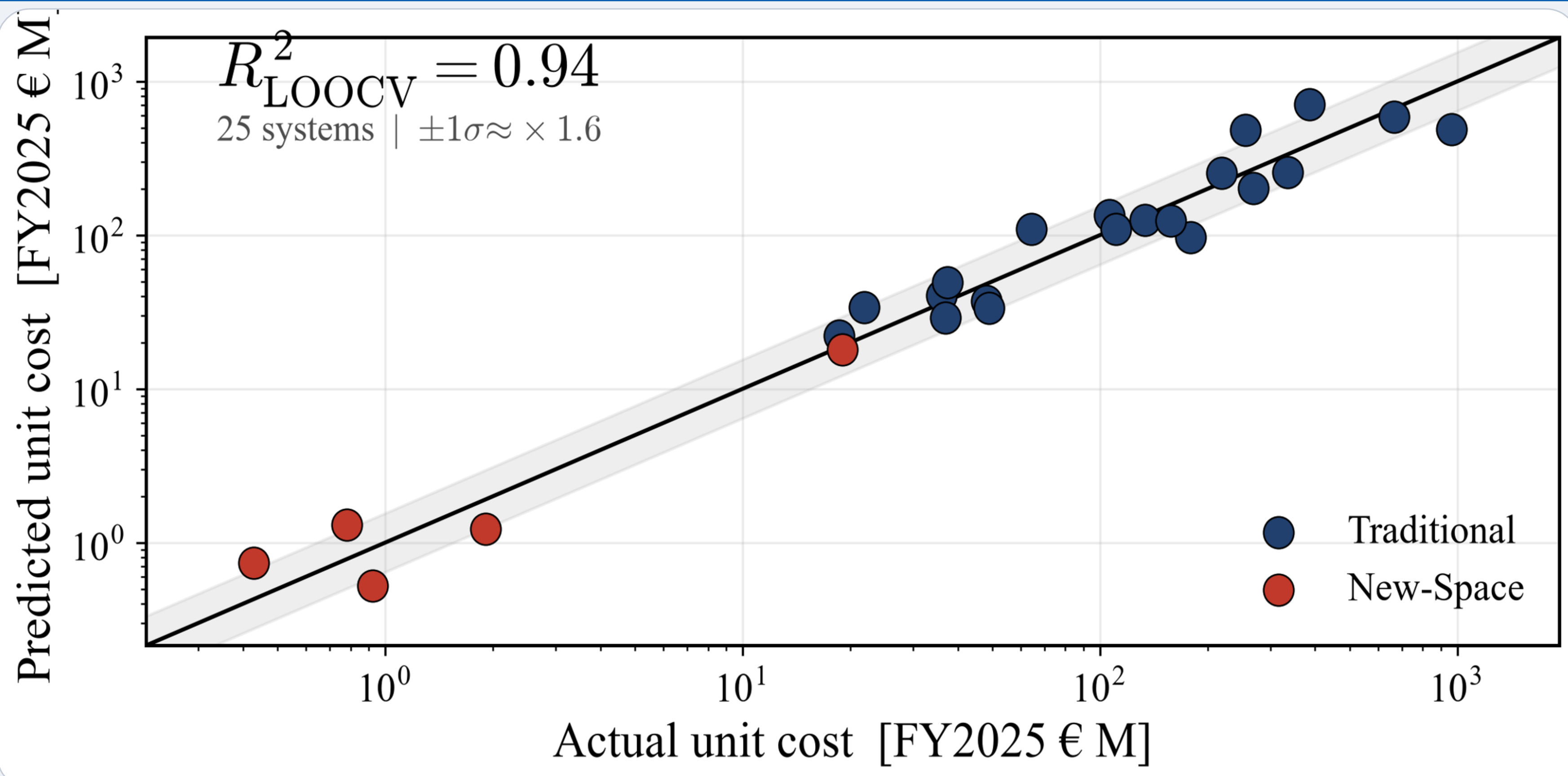
Iso-cost contours of total program cost TOTAL(M,N). VPL Traditional tracks the SMAD benchmark closely, with VPL New-Space a full cost band below.

Cost-Plus vs New-Space Economy



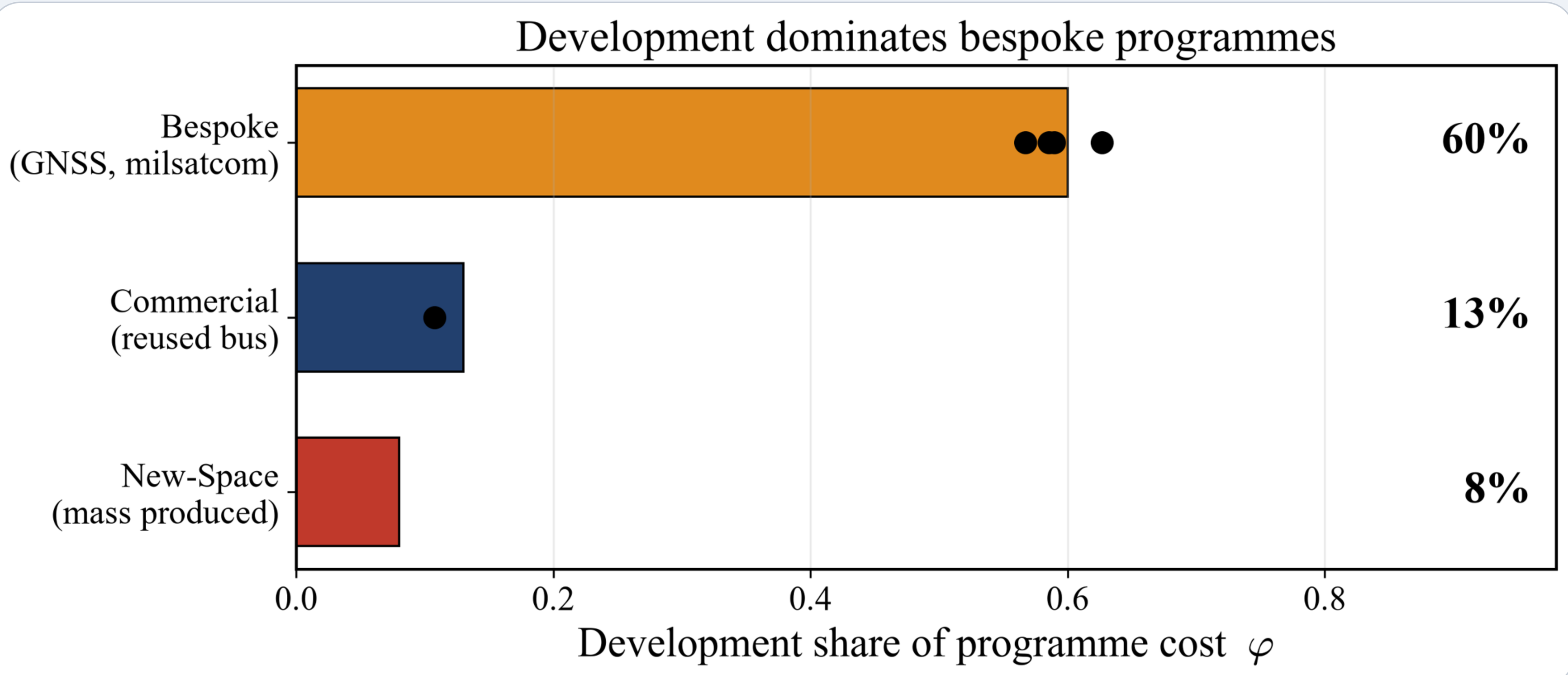
Mass-produced systems are 5–21× cheaper than traditional and bespoke ones.

Validation



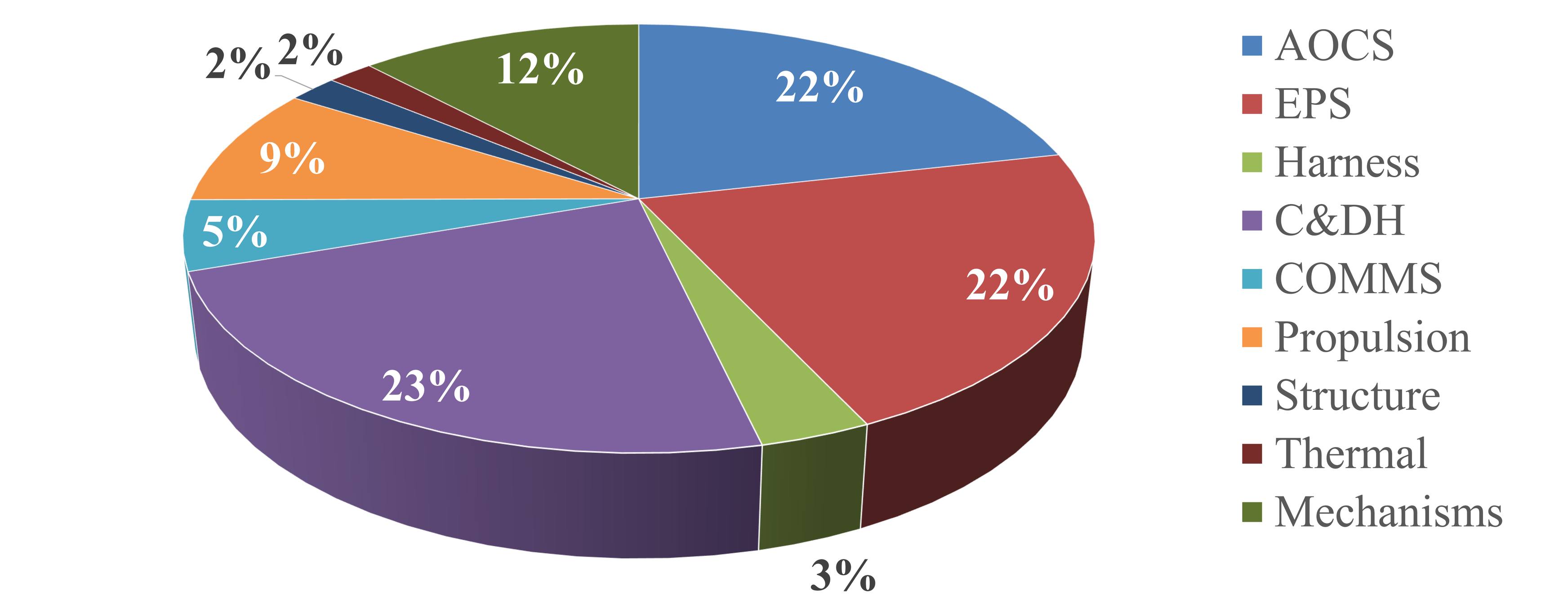
CER tracks cost across 3 orders of magnitude (R^2 = 0.85 for traditional-only).

Development vs production



The DEV/PFM split fixed by disclosed programme budgets (US SARs, ESA, GAO).

Planetary cost factor



Total planetary factor : 7.2% ± 4.5% (Mars bus-cost increase)

Data and code

GitHub repository with the dataset, per-number sources, the Python model, the reproduction notebook, this poster and a detailed report.
github.com/cholevasm/vpl-cost-model

Author Contact

Minas Cholevas

Giannis Vardanikas